

Shahrooz Pouryousef

Gothenburg, Sweden | | [Homepage](#) | [Google Scholar](#) | [LinkedIn](#)

U.S. Permanent Resident

Research Interests

Network security, AI systems security, and distributed quantum computing

Appointments

Postdoctoral Researcher

Chalmers University of Technology

2026 – Present
Gothenburg, Sweden

Research Scientist

Cisco Research

2025
Santa Monica, CA

Research Scientist Intern

Cisco Research

Summer 2023
Santa Monica, CA

Machine Learning Research Intern

ByteDance AI Lab

Summer 2021

Graduate Research Assistant

University of Massachusetts Amherst

2021 – 2025
Amherst, MA

Research Assistant

University of Massachusetts Amherst

2017 – 2021
Amherst, MA

Education

Ph.D. in Computer Science

University of Massachusetts Amherst, Amherst, MA
Advisor: Don Towsley
Thesis: Resource Allocation in Quantum Networks

2021 – 2025

M.S. in Computer Science

University of Massachusetts Amherst, Amherst, MA
Advisors: Arun Vencataramani and Lixin Gao
Research: Interdomain Routing and Internet-Scale Measurement of Online Information Ecosystems

2017 – 2021

M.S. in Computer Engineering

Sharif University of Technology, Tehran, Iran
Research: Security and Privacy in Smartphone Applications

2013 – 2015

B.S. in Information Technology

University of Tabriz, Tabriz, Iran

2009 – 2013

Selected Publications ([Google Scholar](#))

Preprints and In-Review

- P1.** Jesus Lopez, **Shahrooz Pouryousef**, Mohammad Saidur Rahman, “Feature Engineering and Traffic Analysis of LLM Agent Traffic,” *under submission to IEEE INFOCOM, 2027.*

- P2.** Eneet Kaur, **Shahrooz Pouryousef**, Nitish Kumar Chandra, Hassan Shapourian, Jiapeng Zhao, Ramana Kompella, Reza Nejabati, “Impact of Network Constraints on Fault-Tolerant Distributed Quantum Computing,” *submitted to IEEE Transactions on Quantum Engineering (TQE)*, 2026.
- P3.** **Shahrooz Pouryousef**, Ilias, Muoi Tran, “Crunching Public BGP Data for Hijack Detection Isn’t a Good Idea,” *Preprint*, 2026.
- P4.** **Shahrooz Pouryousef**, Don Towsley, “Routing to Ruin: Exploiting SWAP Paths for Circuit Fingerprinting in Distributed Quantum Machine Learning,” *Preprint*, 2026.
- P5.** **Shahrooz Pouryousef**, Lixin Gao, Don Towsley, “Robust Path Selection in Software-Defined WANs Using Deep Reinforcement Learning,” *arXiv preprint*, 2022.

Journal Articles

- J1.** Eneet Kaur, **Shahrooz Pouryousef**, Hassan Shapourian, Jiapeng Zhao, Michael Kilzer, Ramana Kompella, Reza Nejabati, “Optimized Quantum Circuit Partitioning Across Multiple Quantum Processors,” *IEEE Transactions on Quantum Engineering (TQE)* 2025.
- J2.** Leonardo Bacciotini, Matheus Guedes De Andrade, **Shahrooz Pouryousef**, Emily A. Van Milligen, Aparimit Chandra, Nitish K. Panigrahy, Nageswara S. V. Rao, Gayane Vardoyan, Don Towsley, “Leveraging Internet Principles to Build a Quantum Network,” *IEEE Network* 2025.
- J3.** **Shahrooz Pouryousef**, Hassan Shapourian, Alireza Shabani, Ramana Kompella, Don Towsley, “Resource Placement for Rate and Fidelity Maximization in Quantum Networks,” *IEEE Transactions on Quantum Engineering (TQE)* 2024.

Conference Proceedings

- C1.** **Shahrooz Pouryousef**, Eneet Kaur, Don Towsley, Reza Nejabati, “Efficient Entanglement Scheduling in Quantum Data Centers,” *SPIE Quantum Computing, Communication, and Simulation VI* 2026.
- C2.** Saeefa Rubaiyet Nowmi, Jesus Lopez, Md Mahmudul Alam Imon, **Shahrooz Pouryousef**, Mohammad Saidur Rahman, “Critical Evaluation of Quantum Machine Learning for Adversarial Robustness,” *IEEE Symposium on Security and Privacy (S&P)* 2026.
- C3.** **Shahrooz Pouryousef**, Lixin Gao, Arun Venkataramani, “Towards Logically Centralized Interdomain Routing,” *USENIX Symposium on Networked Systems Design and Implementation (NSDI)* 2020.
- C4.** **Shahrooz Pouryousef**, Ali MontazerAlghaem, “Do LLMs Understand Collaborative Signals? Diagnosis and Repair,” *ACM Conference on Human Information Interaction and Retrieval (CHIIR)* 2026.
- C5.** **Shahrooz Pouryousef**, Hassan Shapourian, Don Towsley, “Minimal Protocols for Entanglement Distribution with Finite Memory Coherence Time,” *IEEE International Conference on Quantum Communications, Networking, and Computing (QCNC)* 2024.
- C6.** Mahdi Chehimi, **Shahrooz Pouryousef**, Nitish K. Panigrahy, Don Towsley, Walid Saad, “Scaling Limits of Quantum Repeater Networks,” *IEEE International Conference on Quantum Computing and Engineering (QCE)* 2023.
- C7.** Leonardo Bacciotini, Aparimit Chandra, Matheus Guedes De Andrade, Nitish K. Panigrahy, **Shahrooz Pouryousef**, Nageswara S. V. Rao, Emily Van Milligen, Gayane Vardoyan, Don Towsley, “A Packet-Switched Architecture for Two-Way Quantum Networks,” *CLEO: Science and Innovations* 2025.
- C8.** **Shahrooz Pouryousef**, Nitish K. Panigrahy, Don Towsley, “A Quantum Overlay Network for Efficient Entanglement Distribution,” *IEEE Conference on Computer Communications*

(INFOCOM) 2023.

- C9. Shahrooz Pouryousef**, Muhammad Daniyal Dar, Suleman Ahmad, Phillipa Gill, Rishab Nithyanand, “Extortion or Expansion? An Investigation into the Costs and Consequences of ICANN’s gTLD Experiments,” *Passive and Active Measurement Conference (PAM) 2020*.
- C10.** Zhaoyu Gao, Ahmad Sepahi, **Shahrooz Pouryousef**, Lu Zhou, Haojin Zhu, “Tradeoff Between Privacy and Utility for Location-Based Recommendation Services,” *IEEE International Conference on Communications (ICC) 2022*.
- C11. Shahrooz Pouryousef**, Mariam Rezaiee, Ata Chizari, “Let Me Join Two Worlds! Analyzing the Integration of Web and Native Technologies in Hybrid Mobile Apps,” *IEEE TrustCom/BigDataSE 2018*.

Workshops & Posters

- WP1.** Saeefa Rubaiyet Nowmi, Jesus Lopez, Md Mahmudul Alam Imon, Viviana Cadena, **Shahrooz Pouryousef**, Mohammad Saidur Rahman, “Towards Robust Quantum Machine Learning,” *IEEE International Conference on Quantum Computing and Engineering (QCE) 2025 Poster*.
- WP2.** Nitish K. Panigrahy, Matheus Guedes De Andrade, **Shahrooz Pouryousef**, Don Towsley, Leandros Tassioulas, “Scalable Multipartite Entanglement Distribution in Quantum Networks,” *IEEE International Conference on Quantum Computing and Engineering (QCE) 2023 Poster*.
- WP3. Shahrooz Pouryousef**, Nitish K. Panigrahy, Monimoy Deb Purkayastha, Sabyasachi Mukhopadhyay, Gert Grammel, Domenico Di Mola, Don Towsley, “Resource Management in Quantum Virtual Private Networks,” *IEEE International Conference on Quantum Computing and Engineering (QCE) 2023 Poster*.
- WP4. Shahrooz Pouryousef**, Hassan Shapourian, Alireza Shabani, Don Towsley, “Quantum Network Planning for Utility Maximization,” *Workshop on Quantum Networks and Distributed Quantum Computing (QN-DQC) 2023*.

Patents

- USPT1. Shahrooz Pouryousef**, Eneet Kaur, Hassan Shapourian, Reza Nejabati, “Network-Aware Scheduling for Distributed Quantum Computing,” *App: US64081440 (Pending)* .
- USPT2. Shahrooz Pouryousef**, Hassan Shapourian, Alireza Shabani, “Sequential Entanglement Swapping in Quantum Networks,” *Pending* .

Teaching Experience

Instructor, COMPSCI 110: Introduction to Programming	Fall 2024
Manning College of Information and Computer Sciences, University of Massachusetts Amherst	Amherst, MA
Instructor, First-Year Seminar	Fall 2023
Manning College of Information and Computer Sciences, University of Massachusetts Amherst	Amherst, MA
Teaching Assistant	2021–2025
Manning College of Information and Computer Sciences, University of Massachusetts Amherst	Amherst, MA
Teaching Assistant	2014–2015
Department of Computer Engineering, Sharif University of Technology Courses: Advanced Computer Networks, Network Security.	Tehran, Iran

Student Advising and Mentoring

Chalmers University of Technology

- **Ilias Chanis** — PhD Student, Computer Science (2026–Present)

Stockholm University

- **Sohini Chakraborty** — Master’s Student, Computer Science (2026)

University of Texas at El Paso (UTEP)

- **Saeefa Rubaiyet Nowmi** — PhD Student, Computer Science
- **Jesus Lopez** — M.S. Student, Computer Science

University of Massachusetts Amherst (CICS)

- **Adhiraj Chadha** — Undergraduate Research Volunteer (URV), 2023
- **B. Hamilton** — Undergraduate Research Volunteer (URV)
- **S. Kale** — Undergraduate Research Volunteer (URV)

Collaborations

- CB1. **Don Towsley**, University of Massachusetts Amherst, MA
- CB2. **Ramana Kompella**, Cisco Quantum Labs, CA
- CB3. **Phillipa Gill**, Google
- CB4. **Hassan Shapourian**, Zyphra Technologies Inc, San Francisco, CA
- CB5. **Mohammad Saidur Rahman**, University of Texas at El Paso
- CB6. **Reza Nejabati**, Cisco Quantum Labs, CA
- CB7. **Eneet Kaur**, Cisco Quantum Labs, CA
- CB8. **Nitish K. Panigrahy**, SUNY Binghamton, NY
- CB9. **Walid Saad**, Virginia Tech, VA
- CB10. **Lixin Gao**, University of Massachusetts Amherst, MA
- CB11. **Ali MontazerAlghaem**, Google, Mountain View, CA
- CB12. **Muoi Tran**, Chalmers University of Technology, Sweden

Professional and Service Activities

- S1. Journal Reviewer.** IEEE/ACM Transactions on Networking (ToN), IEEE Transactions on Quantum Engineering (TQE), IEEE Network.
- S2. Conference Reviewer.** S&P and IEEE International Conference on Communications (ICC), 2026 and 2023.
- S3. Graduate Student Representative.** CICS, University of Massachusetts Amherst, 2022–2023.
- S4. Industry Initiative Officer.** Student Leadership Council (SLC), Center for Quantum Networks (CQN), 2023–2024.
- S5. Social Committee Member.** Manning College of Information and Computer Sciences (CICS), University of Massachusetts Amherst, 2019–2020.
- S6. Workshop Organizer.** Quantum Security and Privacy (QSec) Workshop, NeurIPS 2026.

- S7. Organizer.** Secure and Trustworthy Quantum Machine Learning (SaTQuML) Workshop, 2026.
- S8. Mentor.** Mentored undergraduate researchers on distributed quantum computing, quantum error correction, and machine learning projects, 2022–2025.